3-Tier System Architecture Design

Student: Sourabh Shelke | Roll No: 23102C0029 | Class: BE CMPN (Computer Engineering)

1. Architectural Overview

The portal is designed as a cloud-native 3-tier microservice:

• **Tier 1: Presentation Layer:** Single-page responsive web dashboard (index.html) with CSS custom properties, KPI summary cards, filter bars, and underwriter audit modals.

• **Tier 2: Business & Application Layer:** Spring Boot 3.2 REST controllers (DocumentApiController) providing validation, heuristic fraud scoring, and workflow state management.

• **Tier 3: Persistence Layer:** Spring Data JPA repository backed by an embedded H2 relational database ensuring zero-dependency local execution.

• **Tier 4: DevOps & Infrastructure Layer:** Production Alpine JRE Docker container (port 8085), Jenkins CI/CD pipeline (port 8080), and Ansible configuration playbooks.